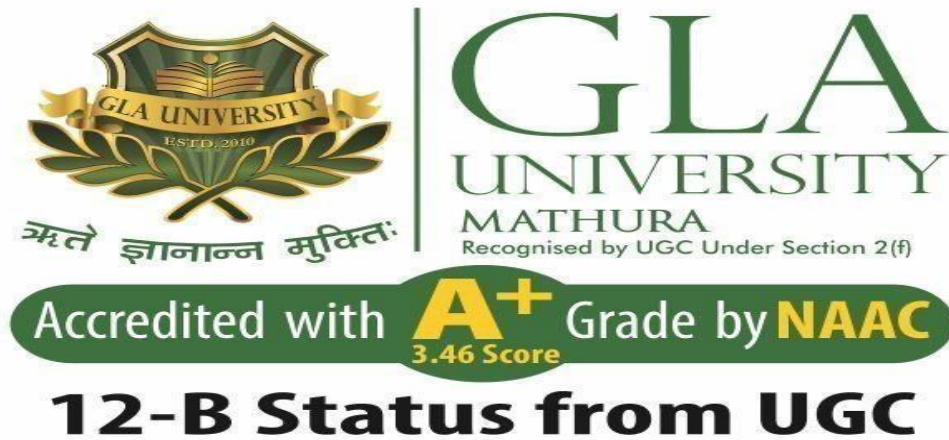


B.Tech CSE (III YEAR – VI SEM) (2025-26)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

17km Stone, NH-2, Mathura-Delhi Road, P.O. Chaumuhan, Mathura – 281406
(Uttar Pradesh) India

Project Title: AI-Powered Drug–Drug Interaction Checker using Graph Neural Networks

Team Leader: Abhigyan Khare UR: 2415500010

Team Member 1: Oshiva singh UR: 2415500322

Team Member 2: Bhavishya Chaudhary UR: 2415500129

Mentor Name: Sir Shivanshu Upadhyay

Signature :

Project Synopsis:

AI-Powered Drug–Drug Interaction Checker using Graph Neural Networks

Cover

- **Project title:** AI-Powered Drug–Drug Interaction Checker using Graph Neural Networks
 - **Team name & ID:** Neural Cyphers(T-100)
 - **Institute / Course:** GLA University, Mini-Project
 - **Version:** v1.0
 - **Date:** 02 september 2026
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1. Overview

- **Problem statement:** Taking multiple medicines at the same time can increase the risk of drug–drug interactions. Manually checking a large number of possible drug combinations can be time-consuming, and some newly documented or under-reported interactions may be difficult to identify.
 - **Goal:** Develop an AI-powered Drug–Drug Interaction Checker that can identify known drug interactions and explore potential interactions using Graph Neural Networks (GNNs).
 - **Non-goals:** Autonomous prescribing, automatic e-prescription generation, food–drug interactions, drug–disease interactions, patient-specific genomic personalization, and replacing professional clinical judgment in v1.
 - **Value proposition:** A graph-based AI decision-support system that combines drug interaction data and molecular information to provide interaction severity and simple, human-readable explanations for healthcare professionals.
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2. Scope and Control

2.1 In-scope

- **Drug search and selection:** Select medicines to check for possible interactions.
- **Known interaction checking:** Check medicines for already known drug interactions.
- **Drug graph creation:** Connect drugs based on their known interactions.
- **GNN-based prediction:** Find possible new or under-reported interactions.
- **Severity and explanation:** Show the severity and reason for an interaction.

2.2 Out-of-scope

- Autonomous prescribing
- Food drug and drug disease interactions
- Patient specific or genetic analysis
- Mobile apps and hospital system integration

2.3 Assumptions

- Drug interaction data is available
- Drug information is reliable
- Molecular information is available

2.4 Constraints

- Limited or incomplete drug data
- Limited computational resources
- GNN prediction errors

2.5 Dependencies

- Drug interaction datasets such as DrugBank, openFDA and TWOSIDES
- RDKit for molecular feature generation
- PyTorch and PyTorch Geometric for GNN model development
- FastAPI and React for the proposed application

2.6 Acceptance criteria and sign-off

- GIVEN a user selects medicines WHEN the check is submitted THEN known drug interactions are displayed
- GIVEN an interaction is identified WHEN the result is displayed THEN severity and a simple explanation are provided
- GIVEN a drug pair has no known interaction WHEN the GNN analysis is performed THEN the system provides a prediction if supported by the model
- System should handle invalid or unavailable drug inputs properly

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Mr. Shivanshu upadhyay	Mentor	Scope, final acceptance		09 september 2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Abhigyan Khare	Abhigyan Khare	Mentor	Team
Research & Data Collection	Abhigyan Khare & Oshiva Singh	Abhigyan Khare & Oshiva Singh	Mentor	Team

Design	Oshiva Singh	Oshiva Singh	Mentor	Team
GNN Model Development	Abhigyan Khare	Abhigyan Khare	Mentor	Team
Implementation	Oshiva Singh & Bhavishya	Oshiva Singh & Bhavishya	Mentor	Team
Testing	Bhavishya Chaudhary	Bhavishya Chaudhary	Mentor	Team
Documentation	Bhavishya Chaudhary	Bhavishya Chaudhary	Mentor	Team

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Abhigyan Khare	Research & Data Lead	Literature review, drug interaction datasets, data collection	Python, Pandas, Research	8 hrs/wk	Abhigyan.khare_cs.aiml24@gla.ac.in
Abhigyan Khare & Oshiva Singh	ML & GNN Lead	GNN model, graph construction, model training	Python, PyTorch, GNN	10 hrs/wk	oshiva.singh_cs.aiml24@gla.ac.in
Oshiva & Bhavishya	Backend & Frontend Developer	API development, database, web interface	FastAPI, React, SQL	8hrs/wk	oshiva.singh_cs.aiml24@gla.ac.in
Bhavishya Chaudhary	Testing & Documentation	Testing, evaluation, documentation and presentation	Python, Testing, Documentation	8 hrs/wk	Abhigyan.khare_cs.aiml24@gla.ac.in

5. Week-wise Plan and Assignments

Weeks	Dates	Milestones	Abhigyan (Lead)	Abhigyan (Backend)	Oshiva (Frontend)	Bhavishya (Testing/ Docs)	Deliverables	Status
1	16–22 Aug	Project Research	Study DDI and finalize idea	Study backend requirements	Study UI requirements	Study project requirements	Research notes	Completed
2	23–29 Aug	Literature Review	Study GNN approaches	Study available APIs	Study existing interfaces	Study existing systems	Literature review	Planned
3	30 Aug–5 Sep	Dataset Research	Study DrugBank, openFDA, TWOSIDES	Plan database	Plan drug search UI	Check dataset quality	Dataset plan	Planned
4	6–12 Sep	System Planning	Finalize GNN approach	Design backend architecture	Design UI	Prepare test plan	System design	Planned
5	13–19 Sep	Data Collection	Collect drug interaction data	Set up backend	Create basic UI	Validate data	Initial dataset	Planned
6	20–26 Sep	Data Preprocessing	Clean and prepare data	Create APIs	Develop drug search	Test processed data	Preprocessed dataset	Planned
7	27 Sep–3 Oct	Graph & GNN Development	Create graph and start GNN model	Integrate model API	Create result interface	Test model output	Initial GNN model	Planned
8	4–10 Oct	Prototype & Evaluation	Evaluate initial model	Complete integration	Complete UI	Testing and documentation	Initial prototype	Planned

6. Users and UX

6.1 Personas

- Doctor/Healthcare Professional: Needs quick and clear information about possible drug interactions.
- Pharmacist: Needs an easy way to check multiple medicines before dispensing.

6.2 Top User Journeys

User → Search drugs → Select medicines → Check interactions → View severity → Read explanation

6.3 User Stories

As a healthcare professional, I want to check multiple medicines together so that I can quickly identify possible drug interactions.

6.4 Accessibility & Localization

- Simple language explanations
- Easy-to-use drug search
- Simple and readable interface

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
DrugBank	Drug Interaction Checker	Doctors, pharmacists, researchers	Drug interaction checking, severity, interaction mechanism and management information.	Paid / Professional	Large drug database and detailed information	Mainly focused on documented interactions	GNN-based exploration of potential under-reported interactions
Drugs.com	Drug Interaction Checker	Patients, doctors, pharmacists	Multiple drug checking, severity levels and interaction explanations	Free	Easy to use and supports many medicines	Mainly database-based checking	Graph-based AI analysis along with known interaction checking
Medscape	Drug Interaction Checker	Doctors and healthcare professionals	Multiple drug checking, interaction severity and mechanisms	Free/ Professional services	Clinician-focused and supports multiple medicines	Mainly focuses on existing interaction information	GNN-based prediction and graph representation of drug relationships

7.2 Positioning

PharmGraph AI focuses on AI-based drug interaction checking using Graph Neural Networks (GNNs).

Measurable Delta:

- **Known interaction checking:** Quickly checks selected medicines using established drug interaction data.
- **AI-based prediction:** Explores potential or under-reported drug interactions using GNN analysis.

- **Explainability:** Shows interaction severity and simple explanations for the identified risk.
- **Graph-based approach:** Combines drug interaction data with molecular and graph-based drug relationships.

8. Objectives and Success Metrics

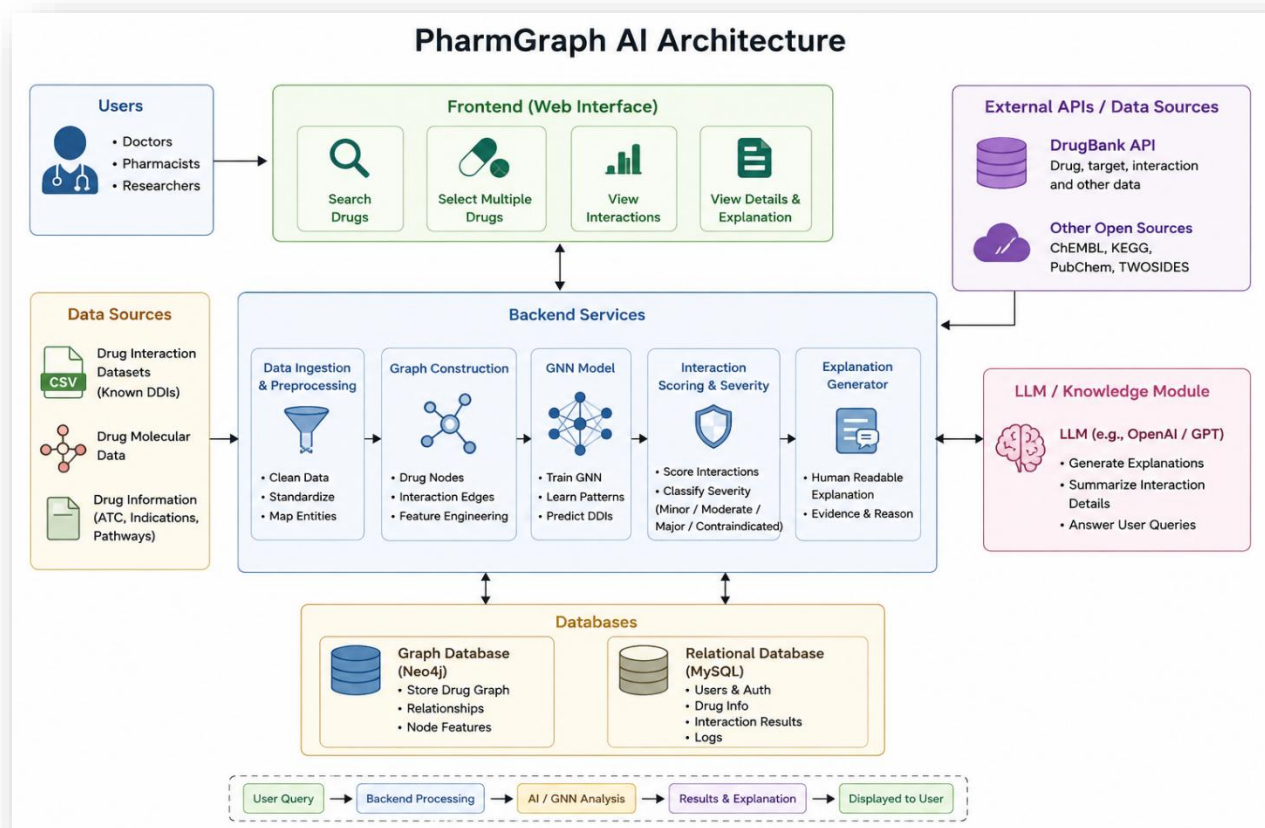
- **O1: Known DDI Detection:** Identify known drug interactions correctly from the selected drug interaction datasets. **(KPI: detection accuracy)**
 - **O2: GNN Prediction:** Explore potential or under-reported drug interactions using the GNN model. **(KPI: prediction performance)**
 - **O3: Severity Classification:** Classify identified interactions into Minor, Moderate, Major, or Contraindicated categories. **(KPI: classification accuracy)**
 - **O4: Explainability:** Provide clear severity information and human-readable explanations for identified interactions. **(KPI: explanation clarity)**
-

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Drug Search and Selection	Allows users to search and select multiple medicines for interaction checking.	Must	Drug Database	GIVEN a user searches for a medicine WHEN a valid drug is selected THEN the drug is added for interaction checking.

Known Interaction Checking	Checks selected medicines against established drug interaction data.	Must	Interaction Datasets	GIVEN multiple drugs are selected WHEN the user checks interactions THEN known interactions are displayed.
GNN-Based Prediction	Uses Graph Neural Networks to explore potential or under-reported drug interactions.	Must	Drug Graph, GNN Model	GIVEN a drug pair has no known interaction WHEN GNN analysis is performed THEN a prediction is provided if supported by the model.
Severity Classification	Classifies drug interactions as Minor, Moderate, Major, or Contraindicated..	Must	Interaction Data, GNN Model	GIVEN an interaction is identified WHEN the result is generated THEN its severity level is displayed.
Interaction Explanation	Provides simple and human-readable information about the identified interaction and its possible risk.	Should	Interaction Analysis	GIVEN an interaction is identified WHEN the result is displayed THEN an understandable explanation is provided.
Interaction Results Dashboard	Displays selected drugs, interaction results, severity, and explanations in a clear interface.	Should	Frontend, Backend	GIVEN the interaction check is completed WHEN results are returned THEN they are displayed clearly to the user.

10. Architecture



10.1 High-Level Architecture

The system follows a modular architecture designed to support drug interaction checking, GNN-based prediction, and clear result presentation.

Clients:

Web-based application developed using React. The client allows users to search and select multiple drugs, check interactions, and view severity and explanations.

- **Backend Services:**

- **Drug Data Service:** Handles drug information, interaction data, and data preprocessing.
- **Graph Construction Service:** Creates the drug interaction graph using drugs as nodes and known interactions as edges.
- **GNN Model Service:** Uses GCN, GraphSAGE, or GAT to analyze drug relationships and predict potential interactions.
- **Interaction Analysis Service:** Calculates interaction probability and classifies severity as Minor, Moderate, Major, or Contraindicated.
- **Explanation Service:** Provides simple, human-readable explanations for the identified drug interactions.
- **Data Stores:**
 - **MySQL/SQLite Database:** Stores drug information, interaction records, prediction results, and system logs.
 - **Graph Database (Neo4j):** Stores drug nodes, interaction relationships, and graph-based features.
- **Integrations:**
 - **Drug Interaction Datasets:** DrugBank, openFDA, and TWOSIDES for drug and interaction data.
 - **RDKit:** Generates molecular features and fingerprints for the GNN model.
 - **GNN Frameworks:** PyTorch and PyTorch Geometric for graph model development.

10.2 Configuration and Secrets

- Environment variables managed using `.env` files
- Sensitive credentials and API keys will not be stored in the source code.

- API keys will be kept secure and used only when required.
 - Access to sensitive configuration will be restricted to authorized team members.
-

11. Data Design

11.1 Core Entities

- Drug
- Drug Interaction
- Molecular Features
- Drug Graph
- Interaction Prediction
- **Severity Classification**

Drug and interaction data will be securely stored with appropriate access control.

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
Drug	drug_id	String	No	Unique ID	Dataset	Unique identifier
Drug	name	String	No	Drug name	Dataset	Name of the drug
Drug	molecular_features	Vector	Yes	Numeric values	RDKit	Used for GNN

Drug Interaction	interaction_id	String	No	Unique ID	Dataset	Unique
Drug Interaction	drug_pair	String	No	Two drug IDs	Dataset	Related drug pair
Drug Interaction	severity	Enum	Yes	Minor/Moderate/Major/Contraindicated	Dataset/Model	Interaction severity
Prediction	probability	Float	Yes	0 to 1	GNN Model	Predicted interaction score

11.2 Schemas and Migrations

- Database schema will be designed according to the required drug and interaction data.
- ER Diagram will be included in the appendix.
- Database migrations will be planned during the implementation phase.

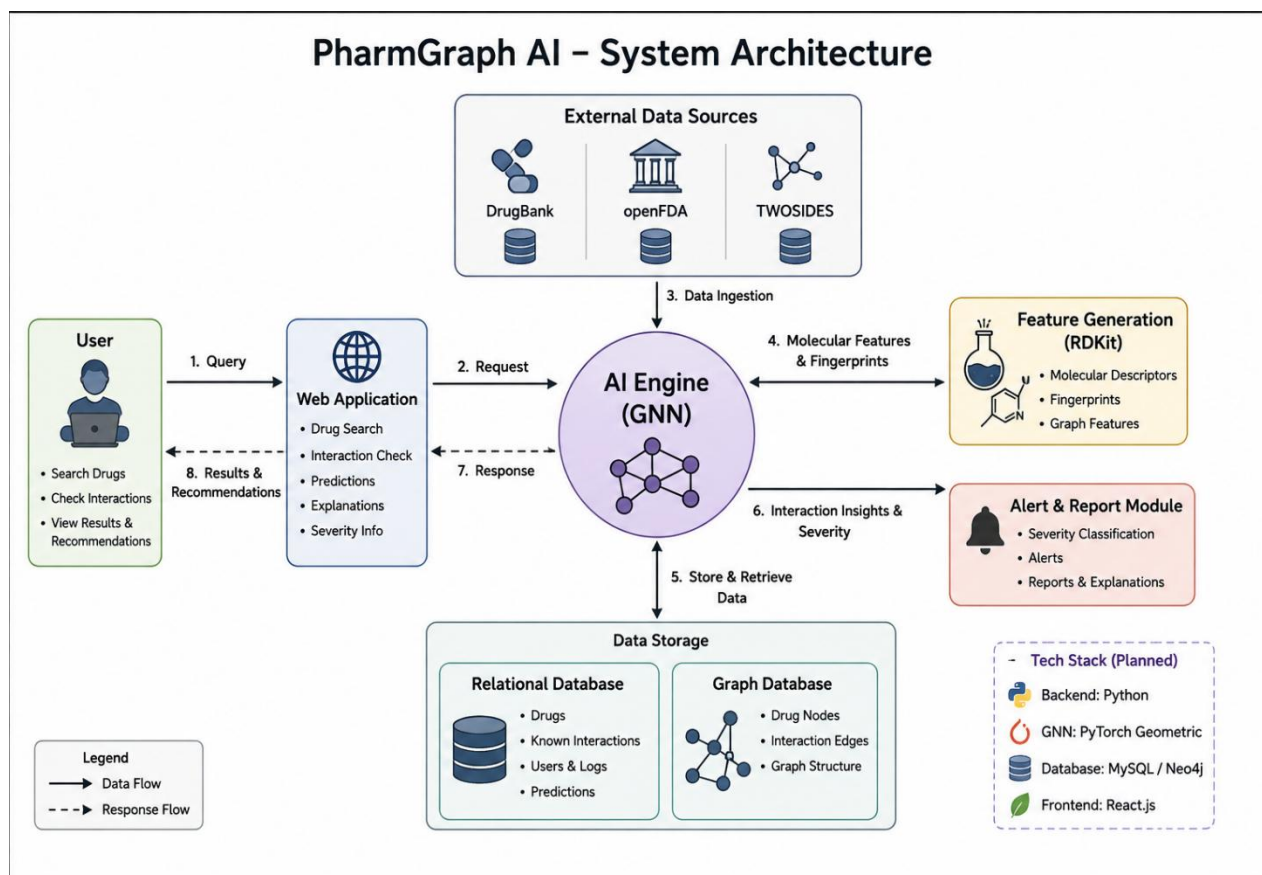
11.3 Privacy, Retention, Backup, and DR

- **User Data:** Only necessary user information will be considered for storage.
- **Drug Data:** Drug and interaction datasets will be stored for system processing and analysis.
- **Data Security:** Stored data will be protected using appropriate access controls.
- **Backup:** Backup procedures will be planned during implementation.
- **Backup:** Backup procedures will be planned during implementation.

12. Technical Workflow Diagrams

The following diagrams will be prepared to explain the proposed system behavior:

- Use Case Diagram
- Sequence Diagram
- GNN Workflow Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- System Architecture Diagram



13. Quality: NFRs and Testing

13.1 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
GNN Model	Model Testing	Python, Scikit-learn	ML Lead	Key scenarios	Acceptable evaluation results
Backend	Unit Testing	Pytest	Backend Lead	70%	No critical defects
UI	Functional Testing	Browser Testing	Frontend Lead	60%	All critical flows pass
API	Integration Testing	Postmann	Backend Lead	All Scenarios	All critical Test pass

13.2 Environments

- Development → Testing → Final Deployment
 - Separate environments will be planned for development and testing
 - Experimental GNN models will be evaluated before final integration.
-

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Drug data	Data tampering	Tampering	Medium	Low	Data Validation	Team
User data	Unauthorized Access	Spoofing	High	Medium	Secure Authentication	Team

14.2 AuthN/AuthZ

- Secure user authentication
- Access control for protected data

14.3 Audit and logging

- Log important system activities and errors.

14.4 Compliance

- Follow institute guidelines and use appropriate datasets.
-

15. Delivery and Operations

15.1 Release plan

- Features will be developed incrementally.
- Planned prototype release at project submission.

15.2 CI/CD and rollback

- **CI Pipeline:** Code check → testing → build → deployment.
- **Version Control:** Project code will be maintained using Git/GitHub.

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
Model prediction error	Above target	ML Lead	Check model and test data
API response time	> 2sec	Backend Lead	Check API and database

15.4 Runbooks

- **Model Error:** Check test results → review model → retrain if required.
- **API Error:** Check logs → identify issue → fix and redeploy.

15.5 Communication plan

- Weekly team meetings and regular progress updates to the mentor.
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16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Limited drug interaction data	Medium	High	12	Use multiple datasets and clean the data	Abhigyan Khare	Open
GNN model prediction errors	Medium	High	12	Evaluate the model using suitable test data	Abhigyan Khare	Open
Limited project time	Hgh	Medium	12	Follow the planned development schedule	Abhigyan Khare	Open

17. Research and Evaluation

- **17.1 Study of Existing Platforms**

Platforms such as DrugBank, Drugs.com, and Medscape provide drug interaction checking and detailed interaction information. However, **PharmGraph AI** will explore a GNN-based approach to identify potential or under-reported interactions along with severity and explanations.

- **17.2 Evaluation Using Historical Data**

PharmGraph AI will be evaluated using existing drug interaction datasets to assess known interaction detection, GNN prediction performance, and severity classification.

- **17.3 User Feedback**

Healthcare-related users or students will be asked to provide feedback on the system's clarity, ease of use, usefulness of explanations, and overall experience.

- **17.4 KPI Tracking**

Healthcare-related users or students will be asked to provide feedback on the system's clarity, ease of use, usefulness of explanations, and overall experience.

18. Appendices

18.1 Glossary

- **AI:** Artificial Intelligence used to perform intelligent analysis and prediction.

- **GNN:** Graph Neural Network used to learn relationships between drugs in a graph.
- **DDI:** Drug-Drug Interaction between two or more medicines.
- **RDKit:** Tool used to generate molecular features and fingerprints.
- **KPI:** Key Performance Indicator used to measure system performance.
- **API:** Application Programming Interface used for communication between software

18.2 References

- DrugBank: <https://go.drugbank.com/>
- openFDA: <https://open.fda.gov/apis/>
- TWOSIDES: <https://tatonetti.c2b2.columbia.edu/>
- RDKit Documentation: <https://www.rdkit.org/docs/>
- PyTorch Geometric: <https://pytorch-geometric.readthedocs.io/>
- FastAPI: <https://fastapi.tiangolo.com/>
- React: <https://react.dev/>
- Scikit-learn: <https://scikit-learn.org/>